New York City Traffic Collisions

New York is one of the most densely populated cities in the world. It has notorious traffic and the NYPD publishes detailed collision records publicly. The dataset included nearly a quarter of a million rows. **238,421 collisions** recorded between **January 2021 and April 2023**.

So, when are collisions happening? The seasonality data showed that summer months are consistently busier, which makes sense as more people out, more cars, more activity. Breaking it down by hour and day of the week revealed a pattern which shows Friday afternoons are brutal. The 3pm to 6pm window across weekdays is when the city is most dangerous, dropping off sharply once the evening sets in.

Then came the question of why. Driver inattention was number one by a wide margin with over 58,000 incidents attributed to it. Road safety is often framed as an infrastructure problem, and sometimes it is, but a lot of it comes down to people simply not paying attention. Failure to yield, following too closely, unsafe speed. The same handful of causes appearing over and over again.

The borough breakdown shows Brooklyn had the most collisions overall, which tracks with it being the most populated borough. But calculating the danger rate (what share of collisions actually resulted in injury or death) changed the picture. Some boroughs with fewer total collisions had a higher proportion of serious ones, which reframes where the real risk sits.

The street-level data also reveals interesting insights. Belt Parkway had 3,728 collisions on its own. One road. To make the ranking more meaningful, a severity score was built that weighted fatalities ten times more heavily than injuries because a street with 500 collisions and 3 deaths is different from one with 500 collisions and none.

Of the 635 people killed across the entire dataset, pedestrians made up the majority and they are not inside or operating any vehicle, they are simply on their own two feet. They were just there. The killed-per-1,000-injured rate makes the gap between pedestrians and motorists stark and hard to look away from. If you're on foot and you get hit, the odds of it being fatal are dramatically higher than if you're in a car. Logically that makes sense.

The further into the data you get, the harder it is to think of it purely as an exercise. There are 635 people in that dataset who didn't make it home. That's what the numbers are.